1.	Explain hierarchical memory organization in computer systems. Discuss inclusion, coherence, and locality properties.
2.	Discuss memory management techniques used in virtual memory systems. Explain paging and segmentation with examples.
3.	Explain coherence problems in shared memory systems. Discuss cache coherence protocols in multiprocessor systems.
4	Explain the concept of Instruction-Level Parallelism (ILP). Discuss various techniques used to increase ILP
5	Compare superscalar, super-pipelined, and VLIW processor architectures with suitable block diagrams.
6	Explain Flynn's taxonomy of parallel architectures with suitable examples.
7	Discuss centralized shared-memory multiprocessor architecture. Explain synchronization techniques used in shared-memory systems.
8	Explain distributed shared-memory architecture in detail. Discuss its advantages, limitations, and applications.
9	Discuss different interconnection networks used in multiprocessor architectures. Compare bus, crossbar, multistage, and hypercube networks.
10	Explain the design of a multicycle RISC processor and discuss how pipelining improves processor performance. Also explain different hazards and their handling techniques.
11	Explain the role of memory hierarchy in improving system performance. Discuss cache coherence, virtual memory, and replacement policies.
12	Discuss the impact of pipeline depth on processor performance. Explain the associated challenges and optimization methods.
13	Design considerations for high-performance processors include pipelining, cache hierarchy, ILP, and multiprocessing. Discuss these aspects in detail with suitable examples.
14	Explain processor organization and architectural overview of modern computer systems. Discuss the role of ALU, control unit, registers, buses, and memory hierarchy.
15	Explain data hazards in pipelined processors with suitable timing diagrams. Discuss forwarding, stalling, and pipeline interlocking techniques.
16	Find ILP in Given Instruction Sequence. Consider the following instructions: I1: R1 = R2 + R3 I2: R4 = R1 + R5 I3: R6 = R7 + R8 I4: R9 = R6 + R10 I5: R11 = R12 + R13 I6: R14 = R1 + R6 a) Identify all data dependencies. b) Draw the dependency graph. c) Find the maximum instruction-level parallelism. d) Arrange the instructions into minimum execution cycles assuming unlimited functional units.
17	Given: I1: A = B + C I2: D = E + F I3: G = A + D I4: H = G + I I5: J = K + L I6: M = J + H Assume each instruction takes one cycle.

	Find: a) Which instructions can execute in parallel? b) Minimum number of cycles required. c) Average ILP.
18	Consider: I1: ADD R1, R2, R3 I2: SUB R4, R1, R5 I3: MUL R6, R4, R7 I4: ADD R8, R2, R3 I5: SUB R9, R8, R6 Identify: a) RAW hazards b) WAR hazards c) WAW hazards d) Instructions that can be reordered safely.

19	Given the following instruction sequence:
	
	`I1: R1 = R2 + R3`
	`I2: R2 = R4 + R5`
	`I3: R6 = R1 + R7`
	`I4: R8 = R2 + R9`
	`I5: R10 = R11 + R12`
	`I6: R13 = R10 + R14`
	
	Find:
	
	a) True dependencies
	b) Anti-dependencies
	c) Output dependencies
	d) Maximum parallel groups.

20	Assume:
	
	• ADD latency = 1 cycle
	• MUL latency = 3 cycles
	• LOAD latency = 2 cycles
	
	Instruction sequence:
	
	`I1: LOAD R1, 0(R2)`
	`I2: ADD R3, R1, R4`
	`I3: MUL R5, R3, R6`
	`I4: ADD R7, R8, R9`
	`I5: MUL R10, R7, R11`
	`I6: ADD R12, R5, R10`
	
	Find:

	a) Data dependency graph b) Number of stalls without scheduling c) Optimized instruction schedule d) Total execution cycles after scheduling.

21	A pipeline has 5 stages with delays:
	
	`S1 = 4 ns`
	`S2 = 6 ns`
	`S3 = 8 ns`
	`S4 = 5 ns`
	`S5 = 7 ns`
	
	Pipeline register delay = 1 ns.
	
	Calculate:
	
	a) Pipeline cycle time
	b) Time to execute 50 instructions
	c) Non-pipelined time for 50 instructions
	d) Speedup.

22	A pipelined processor has ideal CPI = 1. The instruction mix is: • 30% load instructions • 20% branch instructions • 50% ALU instructions Assume: • 40% of loads cause 1 stall cycle • 50% of branches cause 2 stall cycles Find: a) Average stall cycles per instruction b) Effective CPI c) Speedup over non-pipelined processor with CPI = 5.
23	A 5-stage pipeline uses a single memory for both instruction fetch and data access. Load/store instructions are 40% of total instructions. Each load/store causes one structural stall. For 1,000 instructions, find:

	a) Number of structural stalls b) Total cycles c) Effective CPI.
24	Identify Addressing Modes Identify the addressing mode used in each instruction. a) MOV R1, R2 b) MOV R1, #25 c) MOV R1, 2000 d) MOV R1, (R2) e) MOV R1, 20(R2) f) MOV R1, @(R2)+ g) JMP LOOP h) ADD R1, R2, R3
25	Calculate Effective Address. Assume: R1 = 1000 R2 = 2000 Memory[1000] = 5000 Memory[2000] = 7000 Memory[2020] = 9000 PC = 4000 Find the effective address and operand value for: a) MOV R3, (R1) b) MOV R3, 20(R2) c) MOV R3, #50 d) MOV R3, @R1 e) JMP 40(PC)